4	**(a)**	$f = 1/T = 1/0.05 = 20$ Hz $v_o = \varpi x_o = (2\pi f)\, x_o$ $x_o = 0.044/2\pi(20) = 3.5 \times 10^{-4}$ m	M1
	(b)	Cosine shape drawn, maximum at $t = 0$, amplitude 3.5×10^{-4} m	A1 A1
	(c)	(any of the following when the velocity is zero) 0.00s, 0.025s,0.050s or 0.075s □	A1
	(d)	Acceleration of plate (in shm) is proportional to (frequency)^2 or $a = \omega^2 x = (2\pi f)^2 x$.	B1
		As frequency increases, acceleration increases until it is equal to g, the acceleration due to gravity.	M1
		When the sand and plate are both free falling at g, the acceleration due to gravity, there is zero contact force between sand and plate when the vibrating surface.	A1
5	(a)	Equipotential lines are closer together near the cable but further apart away	B1